

LPO 8800: Statistical Methods in Education Research
Vanderbilt University
Midterm Exam
October 8, 2024

Name: _____

By signing below, I agree to the terms of Vanderbilt University's honor code.

Signature: _____

Instructions: Read each question carefully and provide clear, concise responses in the space provided. Be sure to complete every part of every question. Partial credit will be given where appropriate. If you make any assumptions to answer a question, please state those explicitly. You may use a calculator and one sheet of formulas/notes. Good luck!

Question 1. Student loan debt for graduates of Borrow U. is believed to be normally distributed with a mean (μ) of \$18,400 and a standard deviation (σ) of \$4,700. Use this information to answer the following questions. **(27 points)**

- (a) Given the information provided above, approximately what fraction of graduates have *more* than \$25,000 in student loan debt? Provide a numeric answer, and indicate how you arrived at this answer. **(5 points)**
- (b) The institutional research office at Borrow U. plans to take a random sample of 30 graduates and calculate their mean student loan debt ($\bar{x}$). What would you expect the sampling distribution of $\bar{x}$ to look like? That is, what would its expected value, standard error, and distributional shape be (if this can be determined)? Briefly explain your answer. **(5 points)**
- (c) Given your sample of graduates, you calculate a mean student loan debt of \$16,500. Assuming the information above about the population distribution is true, what is the probability of drawing a random sample of $n = 30$ with a mean of \$16,500 *or less*? Briefly justify your answer. **(5 points)**

- (d) If the standard deviation in the population (σ) were \$5,500 rather than \$4,700, would your sample mean of \$16,500 be more or less likely than in part (c)? Briefly explain your answer (no calculation necessary). (**4 points**)
- (e) If your sample were of 40 graduates rather than 30 (and the standard deviation remained at $\sigma = \$4,700$), would your sample mean of \$16,500 be more or less likely than in part (c)? Briefly explain your answer (no calculation necessary). (**4 points**)
- (f) If your sample were back at the original 30 graduates, but the standard deviation were estimated, not known ($s = \$4,700$), would your sample mean of \$16,500 be more or less likely than in part (c)? Briefly explain your answer. (**4 points**)

Question 2 The following questions relate to the Early Childhood Longitudinal Study–Kindergarten Cohort (ECLS-K), a longitudinal study of more than 20,000 children followed from kindergarten through 8th grade. **(30 points)**

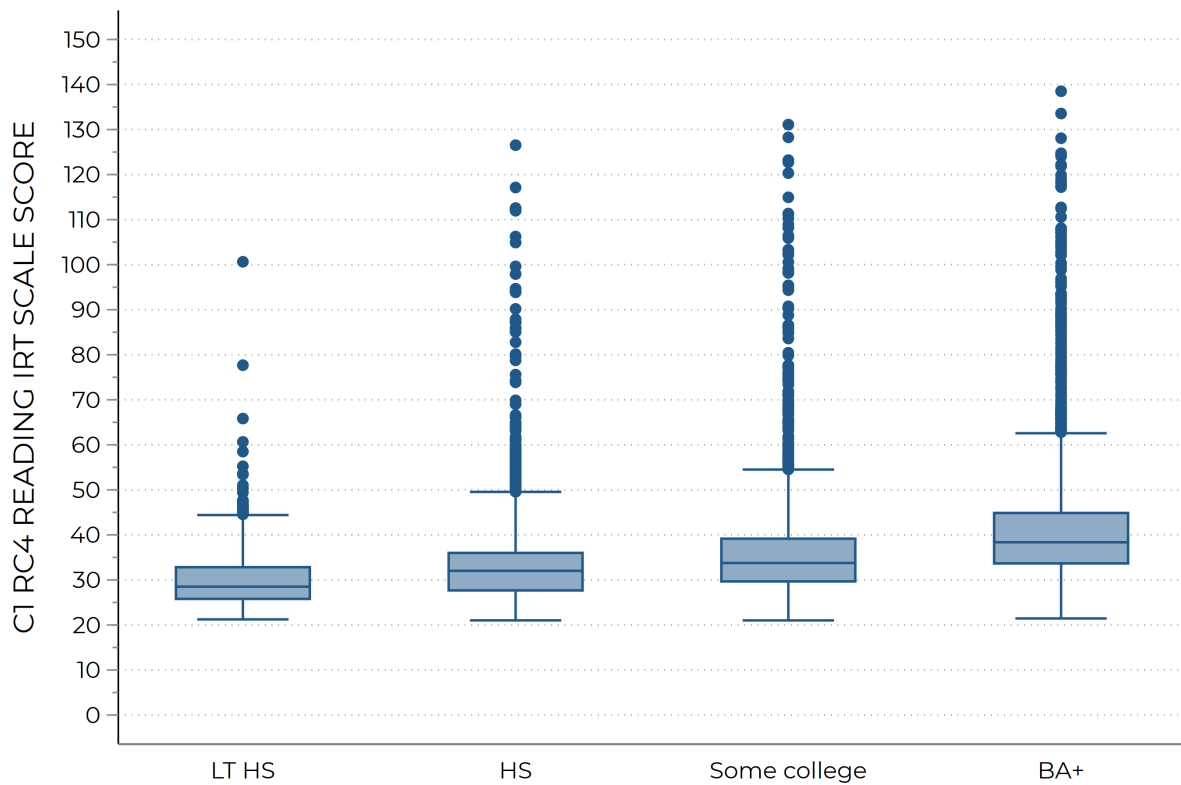
- (a) Consider the frequency distribution for WKMOMED (mother’s education level) below. Why would the mean and standard deviation of this variable not be appropriate statistics to report? Briefly explain. **(4 points)**

```
. fre WKMOMED
```

```
WKMOMED -- WK MOTHER'S EDUCATION LEVEL
```

		Freq.	Percent	Valid	Cum.
Valid	-9 NOT ASCERTAINED	1	0.00	0.00	0.00
	-1 NOT APPLICABLE	330	1.54	1.64	1.64
	1 8TH GRADE OR BELOW	926	4.33	4.60	6.24
	2 9TH - 12TH GRADE	1899	8.87	9.43	15.67
	3 HIGH SCHOOL DIPLOMA/EQUIVALENT	5993	27.99	29.76	45.42
	4 VOC/TECH PROGRAM	1097	5.12	5.45	50.87
	5 SOME COLLEGE	5267	24.60	26.15	77.02
	6 BACHELOR'S DEGREE	3052	14.26	15.15	92.18
	7 GRADUATE/PROFESSIONAL SCHOOL-NO DEGREE	359	1.68	1.78	93.96
	8 MASTER'S DEGREE (MA, MS)	886	4.14	4.40	98.36
	9 DOCTORATE OR PROFESSIONAL DEGREE	331	1.55	1.64	100.00
	Total	20141	94.08	100.00	
Missing	.	1268	5.92		
Total		21409	100.00		

- (b) Below is a box plot showing the distribution of 1st grade reading scores for four categories of maternal education (less than high school, high school diploma, some college, and bachelor’s degree or higher). How would you say the central tendency, variability, and skewness differs across these four categories, if at all? Briefly explain. **(4 points)**



- (c) True or false? A larger sample size would reduce the skewness of the 1st grade reading score distribution (as pictured above for the four maternal education groups) and result in a more normal-shaped distribution. Briefly explain your reasoning. (4 points)

- (d) Use the following Stata output to provide the statistics requested below. The variable here is a measure of self-concept in 7th grade. (Note for purposes of this problem, only a random sample of the full data were used). (**6 points**)

```
. summ C7CONCPT if C7CONCPT>=-9, det
```

C7 SELF CONCEPT				

	Percentiles	Smallest		
1%	-2.03	-3.06		
5%	-1.26	-2.44		
10%	-.9	-2.12	Obs	330
25%	-.45	-2.03	Sum of Wgt.	330
50%	.04		Mean	-.0330303
		Largest	Std. Dev.	.701972
75%	.47	1.02		
90%	.81	1.02	Variance	[omitted]
95%	1.02	1.02	Skewness	[omitted]
99%	1.02	1.02	Kurtosis	4.013447

Find the following:

- Inter-quartile range
- Variance
- Median
- 95% confidence interval for the population mean

- (e) Based on the information provided in part (d), how would you describe the shape/symmetry of the self concept scale distribution? Briefly explain how you arrived at your answer. (**4 points**)
- (f) Your research assistant transformed the variable in part (d) to z -scores and reported back to you the average z -score for select subgroups of students. For example, students in higher-income households had a mean self-concept z -score of 0.85. How would you explain this value in words, and what is the mean self concept for students in higher-income households on the original scale? (**5 points**)
- (g) True or false? The z -score transformation in part (f) will result in a variable with an approximately normal distribution. Briefly explain your reasoning. (**3 points**)

Question 3. Listed below are the joint probabilities that a randomly drawn middle school student has a computer at home (or not) and took an advanced placement (AP) course in high school (or not). For example, the probability a randomly drawn student has a computer at home and takes an advanced placement course in high school is 0.290. Use this information to answer the following questions. **(13 points)**

		Computer at home		Total
		No	Yes	
AP course in HS	No	0.235	0.185	
	Yes	0.290	0.290	
Total				1.000

(a) What is $P(\text{AP course})$? **(3 points)**

(b) What is $P(\text{AP course} \mid \text{computer at home})$? **(5 points)**

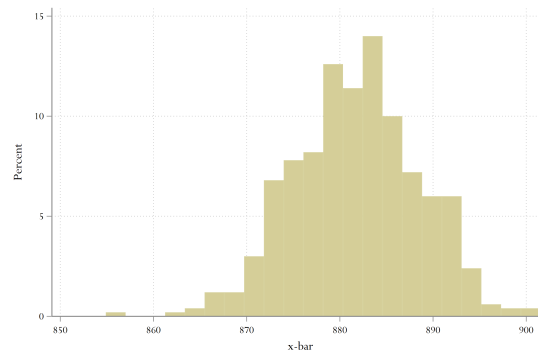
(c) Using the results above, determine whether taking an AP course in high school is independent of having a computer at home, and explain your answer. **(5 points)**

Question 4. The descriptive statistics and histogram below represent a simulated sampling distribution for $\bar{x}$ in which $n = 25$ for each sample from the population. 500 random samples were drawn as part of the simulation. **(10 points)**

```
. summ mean,detail
```

r(mean)				

	Percentiles	Smallest		
1%	555.0983	551.7162		
5%	561.0122	552.4836		
10%	563.9032	553.7741	Obs	500
25%	568.9125	553.777	Sum of Wgt.	500
50%	574.2776		Mean	574.341
		Largest	Std. Dev.	8.039671
75%	579.6404	596.8365		
90%	583.7625	597.5052	Variance	64.63631
95%	588.4326	597.7547	Skewness	.0661987
99%	595.954	599.02	Kurtosis	3.170898

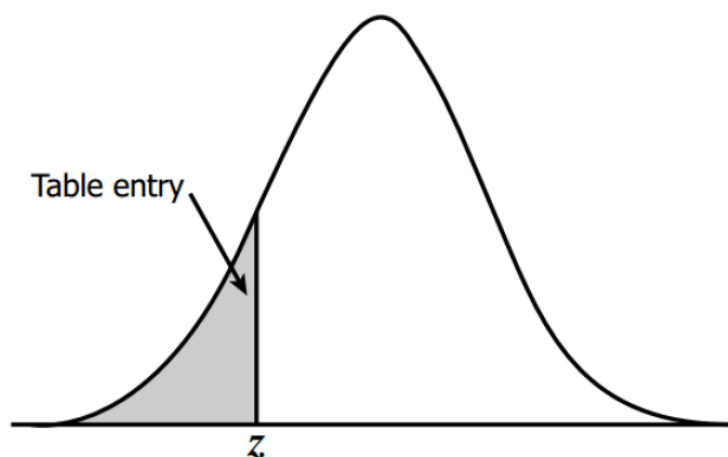


(a) What is the standard error of $\bar{x}$ based on this simulation? Briefly explain your answer. **(5 points)**

(b) It is easy to show that the sample mean $\bar{x}$ is *unbiased*. Briefly explain what this means. **(5 points)**

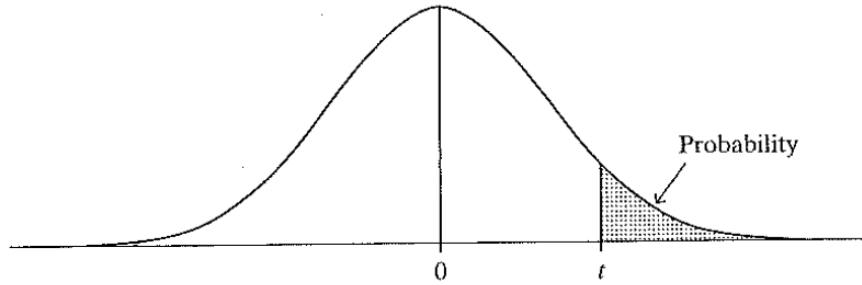
Question 5. In a recent Fox News poll of 512 Republican voters selected at random, 63% reported they think Donald Trump should be the Republican nominee for president in 2024. Provide a 99% confidence interval for the true population proportion of Republican voters who think Donald Trump should be the Republican nominee in 2024. Show your calculation. What is the margin of error? **(10 points)**

Question 6. You wish to survey a random sample of public schoolteachers in Tennessee in order to estimate the population mean number of hours worked per week during the school year. Your best estimate of the population standard deviation—based on similar studies of teachers elsewhere—is 90 minutes. How many teachers will you need to sample in order for your estimate to be within **10** minutes of the population mean 95% of the time? Show how you arrived at your answer. **(10 points)**



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

TABLE B: t Distribution Critical Values



df	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.656	318.289
2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.894
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.611
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195